

INFOSYS SPRINGBOARD VIRTUAL INTERNSHIP

A Project Report

On

SUPPLY CHAIN VISIBILITY SYSTEM WITH OPTIMIZATION ANALYTICS

Milestone - 1 Documentation: Data Modeling & KPI
Foundation

Submitted By-

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1. INTRODUCTION

The **Supply Chain Visibility System with Optimization Analytics** is a data analytics project developed as part of the Infosys Springboard Virtual Internship. The main purpose of the project is to analyze vessel movement and logistics-related data and convert the available raw data into meaningful information through data preparation, modeling, KPI development, and interactive dashboards.

In Milestone 1, the focus was on establishing the basic foundation of the project. The dataset was imported into Power BI, examined to understand its structure and important attributes, and prepared for analysis. Data cleaning and transformation were performed wherever required, followed by the creation of a simple data model and relationship between the required tables. DAX measures were then developed to calculate important KPIs, which were used to create an interactive dashboard for understanding vessel activity, speed, distance, ETA, vessel type, and status.

1.1 MILESTONE 1 REQUIREMENTS

The major requirements completed during Milestone 1 were:

- **Data Import:** Import the provided logistics/vessel movement dataset into Power BI.
- **Data Preparation:** Examine the dataset structure, identify relevant columns, and prepare the data for analysis.
- **Data Cleaning:** Handle data types, unnecessary inconsistencies, and prepare columns required for analysis.
- **Data Modeling:** Create an appropriate data model using the required tables and establish relationships between them.
- **DAX Development:** Create DAX measures and calculated columns required for analytical calculations.
- **KPI Development:** Develop important KPIs such as total unique vessels, total vessel records, average speed, and average ETA.
- **Dashboard Development:** Build an interactive dashboard containing KPI cards, charts, slicers, and a detailed vessel information table.
- **Data Visualization:** Present the analyzed information in a clean and easy-to-understand visual format.

1.2 MILESTONE 1 OBJECTIVE

- The overall objective of Milestone 1 was to build a strong analytical foundation for the Supply Chain Visibility System, so that the prepared data, data model, KPIs, and dashboard can be further extended in the upcoming milestones.

2. DATASET OVERVIEW

The dataset used in this project contains vessel movement and maritime logistics information. It records different observations related to vessels, including their identification details, location, movement characteristics, cargo information, estimated arrival time, vessel type, status, and weather conditions. The dataset provides the basic information required to understand vessel activity and develop supply-chain visibility insights.

The dataset contains a large number of vessel movement records, which makes it suitable for analyzing vessel behavior and operational patterns. Each record represents an observation of a vessel at a particular time, allowing different parameters such as speed, location, distance, ETA, and vessel status to be analyzed together.

2.1 MAJOR CATEGORIES OF DATA

The dataset can be broadly divided into the following categories:

VESSEL DATA CATEGORY BREAKDOWN		
Data Category	Examples of Fields	Purpose
 Vessel Identification	MMSI, VesselName, CallSign, IMO	Identifies individual vessels
 Time Information	BaseDateTime, MovementDate, Hour	Helps analyze vessel movement over time
 Location Information	LAT, LON, destination latitude/longitude	Represents vessel position and destination
 Movement Information	SOG, COG, SOG_kmh, Speed_Category	Describes vessel movement and speed
 Operational Information	Cargo, VesselType, Status	Provides information about vessel activity
 ETA Information	ETA_hours, ETA_min	Helps analyze estimated arrival time
 Weather Information	Visibility, Cloud, Humidity, Pressure, Temperature, Precipitation	Provides environmental conditions associated with vessel movement

Source Data: Standard Maritime Reporting

2.2 IMPORTANT DATASET FIELDS

Some of the important fields used during Milestone 1 are:

- **MMSI:** A unique maritime identification number used to identify a vessel.
- **VesselName:** Name of the vessel.
- **BaseDateTime:** Date and time at which the vessel observation was recorded.
- **LAT:** Latitude of the vessel's recorded position.
- **LON:** Longitude of the vessel's recorded position.
- **SOG:** Speed over ground, representing the vessel's movement speed.
- **SOG_kmh:** Speed converted into kilometers per hour.
- **Speed_Category:** A categorized representation of vessel speed such as *Stopped*, *Slow*, *Moderate*, and *Fast*.
- **Cargo:** Represents the cargo-related value associated with the vessel record.
- **VesselType:** Represents the category/type of vessel.
- **Status:** Represents the operational/status value recorded for the vessel.
- **ETA_hours:** Estimated time of arrival represented in hours.
- **MovementDate:** Date extracted from the vessel movement timestamp for date-based analysis.
- **Hour / Hour_Label:** Additional time fields created to analyze vessel activity and ETA patterns at different hours.
- **Weather Fields:** These include environmental parameters such as visibility, cloud conditions, humidity, pressure, temperature, and precipitation.

2.3 DATASET UNDERSTANDING

During the initial examination of the dataset, we identified that it contains both raw operational information and fields that can be transformed into analytical features. For example, the original timestamp can be used to create date and hour-based fields, while vessel speed can be grouped into meaningful speed categories.

This understanding helped us decide which fields were directly useful for analysis and which additional fields needed to be created during the data preparation stage. The prepared dataset was then used for data modeling, DAX calculations, KPI development, and dashboard creation.

3. TOOLS & TECHNOLOGIES

The development of the Supply Chain Visibility System was carried out mainly using Microsoft Power BI and its integrated data analysis tools. Different components of Power BI were used for importing, preparing, transforming, modeling, calculating, and visualizing the vessel movement data.

Tool / Technology	Purpose in the Project
Microsoft Power BI Desktop	Main platform used to develop the data model, KPIs, visualizations, and interactive dashboard
Power Query	Used for importing, transforming, and preparing the raw dataset before analysis
DAX (Data Analysis Expressions)	Used to create calculated columns and measures for KPI and analytical calculations
Power BI Data Model	Used to organize the tables and establish the required relationship between the date and vessel movement data
Power BI Visualizations	Used to present the analyzed information through KPI cards, charts, slicers, and tables

Table view from [Project Name]

4. DATA IMPORT

The first step in developing the Supply Chain Visibility System was to import the provided vessel movement dataset into Microsoft Power BI Desktop. Since the project involves a large number of vessel movement records, Power BI was selected as the main platform for handling, analyzing, and visualizing the data.

The dataset was imported into Power BI and loaded into the Power Query Editor, where the structure and contents of the data could be examined before creating the final analytical model. During this initial stage, the available columns were reviewed to understand the type of information contained in the dataset and to identify fields that would be useful for further analysis.

4.1 DATASET IMPORT PROCESS

The data import process was carried out through the following steps:

1. Power BI Desktop was opened and a new project was created.
2. The Get Data option was used to connect Power BI with the provided dataset.
3. The required dataset was selected and loaded into Power Query.

4. The available columns and sample records were inspected to understand the structure of the data.
5. Data types of important fields were reviewed, especially fields related to dates, vessel identification, location, speed, cargo, ETA, and vessel type.
6. The imported query was organized and renamed as Fact_VesselMovement to represent the main vessel movement table in the data model.
7. The data was then prepared for the cleaning and transformation stage.

4.2 UNDERSTANDING THE IMPORTED DATA

After importing the dataset, we identified that the main table contained information related to vessel identification, movement, geographical location, cargo, vessel type, status, ETA, and weather conditions.

The BaseDateTime field was particularly important because it contained the date and time associated with vessel observations. This field later helped us create additional time-related fields such as BaseDate, MovementDate, Hour, and Hour_Label, which were required for time-based analysis in the dashboard.

Similarly, fields such as MMSI, SOG, LAT, LON, Cargo, VesselType, Status, and ETA_hours were reviewed because they could directly support the KPIs and visualizations planned for Milestone 1.

4.3 PREPARING THE DATA FOR THE NEXT STAGE

The imported data was not immediately used for dashboard development. First, it was taken through a data preparation and cleaning process to make the information suitable for analysis.

This ensured that the dataset had appropriate data types, useful analytical fields, and a consistent structure before moving to the modeling stage. After completing the initial import and inspection, the next step was therefore to clean and transform the dataset using Power Query.

5. DATA PREPARATION AND CLEANING

After importing the dataset, the next step was to prepare the raw vessel movement data for analysis. Data preparation was important because the raw dataset contained different types of information, and the fields needed to be reviewed and organized before they could be used effectively for calculations and visualizations.

The data preparation process was mainly carried out using Power Query in Power BI. The objective was to make sure that the important fields had appropriate data types, identify fields required for analysis, and create additional fields that could provide more useful analytical information.

5.1 DATA TYPE VALIDATION

The data types of the important columns were reviewed during the preparation stage. Correct data types are important because Power BI uses them differently during calculations, filtering, sorting, and visualization.

The following types of fields were checked:

- **Vessel identification fields** such as MMSI and IMO were treated as identification values rather than numerical measurements.
- **Date and time fields** were reviewed so they could be used for date and time analysis.
- **Latitude and longitude** fields were maintained as numerical values for geographical analysis.
- **Speed-related fields** were checked as numerical values so that averages and categories could be analyzed.
- **ETA and distance fields** were reviewed for numerical calculations.
- **Categorical fields** such as vessel status, vessel type and cargo were prepared for filtering and classification.

5.2 DATA PREPARATION

The following fields were created:

Columns Created During Data Preparation

Column Created	Purpose
BaseDate	Extracts only the date from BaseDateTime for date-based analysis and for connecting the fact table with the DimDate table.
Hour	Extracts the hour from BaseDateTime to support hourly analysis and provide a numeric field for correct chart sorting.
Hour_Label	Converts the hour into a readable format such as 00:00, 01:00 and 02:00 for clear display on dashboard charts.

6. DATA MODELING & RELATIONSHIP CREATION

After completing the data preparation and cleaning process, the next step was to create a structured data model in Power BI. Data modeling was required to organize the prepared data and establish a connection between the vessel movement records and the date information used for time-based analysis.

The model developed for Milestone 1 consists mainly of two tables: **Fact_VesselMovement** and **DimDate**. The **Fact_VesselMovement** table contains the detailed vessel movement records, while the **DimDate** table provides a structured date reference for date-based filtering and analysis.

6.1 FACT_VESSELMOVEMENT TABLE

The **Fact_VesselMovement** table was used as the main table of the data model. It contains the detailed records imported and prepared during the earlier stages.

The table contains information related to:

- Vessel identification
- Date and time of movement
- Vessel location
- Speed and speed category
- Vessel type
- Vessel status
- Cargo information
- Distance
- ETA
- Other operational and weather-related fields

Since multiple movement records can exist for the same vessel and date, this table represents the detailed movement-level data used for calculations and dashboard visualizations.

6.2 DIMDATE TABLE

A separate **DimDate** table was created to provide a dedicated date structure for the project.

Instead of depending completely on the date values stored within the movement table, the date dimension provides a consistent source for date-based analysis and filtering.

The DimDate table was connected to the prepared date field in the Fact_VesselMovement table. This structure also provides a foundation for future time-based analysis as the project is expanded in later milestones.

6.3 RELATIONSHIP CREATION

A **one-to-many (1:*) relationship** was created between the two tables.

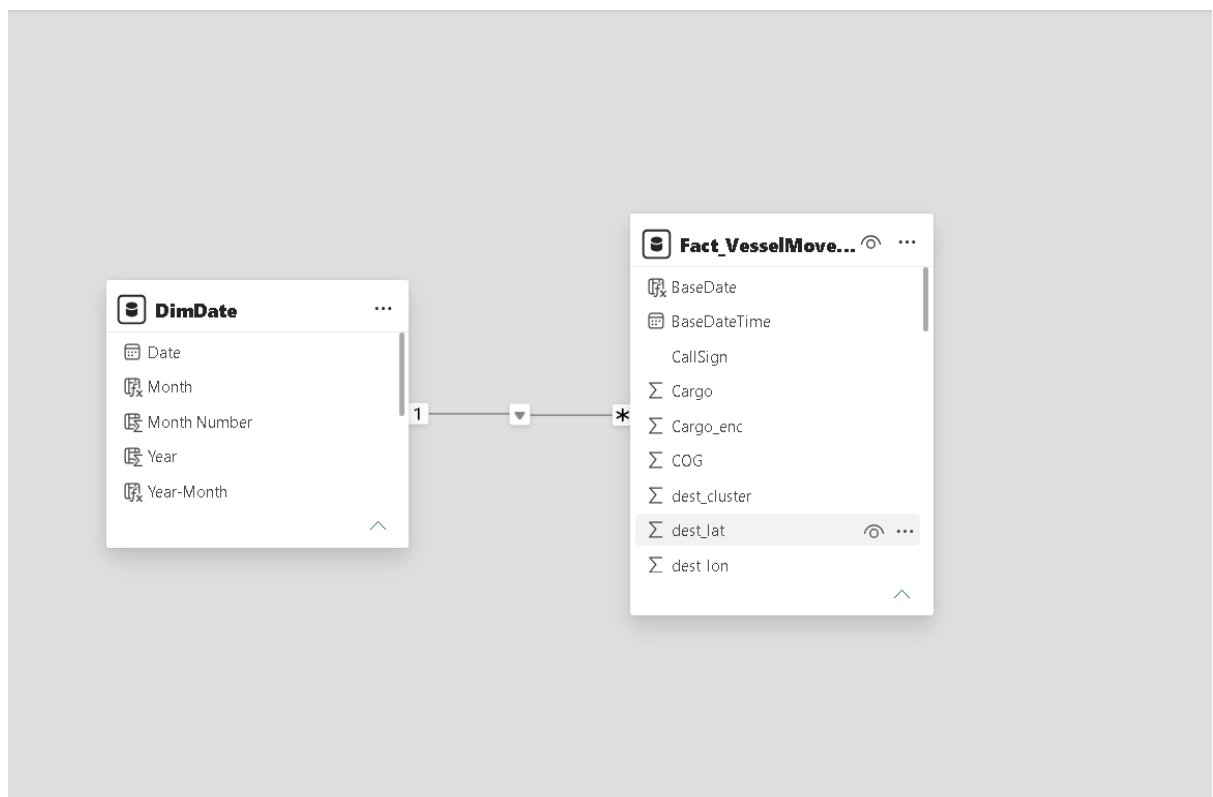
The relationship can be represented as:

DimDate (1) → Fact_VesselMovement (*)

The **DimDate** table is placed on the one side because each date is represented once in the date dimension. On the other hand, the **Fact_VesselMovement** table is placed on the many side because multiple vessel movement records can belong to the same date.

The relationship was created using the appropriate date fields from both tables. It was kept active so that date-based selections and filters could be applied to the vessel movement data.

This relationship allows the dashboard to use the date dimension for analysis while the detailed movement records remain in the main fact table.



7. DAX DEVELOPMENT

After creating the data model, the next step was to develop the calculations required for analysis. Data Analysis Expressions (DAX) was used in Power BI to create calculated columns and measures from the prepared vessel movement data.

DAX was important because the raw dataset alone does not directly provide all the values required for the dashboard. By using DAX, the required information could be calculated dynamically and then used in KPI cards and other visualizations.

7.1 DAX CALCULATED COLUMNS

During the data preparation stage, additional time-related columns were created to support the dashboard's time-based analysis.

The main calculated columns were:

- **BaseDate** — extracts the date from BaseDateTime and is used for date-level analysis.
- **Hour** — extracts the hour from the movement timestamp and is used for hourly analysis.
- **Hour_Label** — provides a readable hour representation for displaying hourly information in charts.

7.2 DAX MEASURES

DAX measures were then created to calculate the main numerical indicators required for Milestone 1. The measures were designed to provide summarized information from the vessel movement data. Unlike fixed values, these measures are calculated dynamically based on the underlying data and can respond to filters applied through the dashboard.

The major measures developed were:

Key DAX Measures Used in the Dashboard

Measure	DAX Formula	Purpose
Total Unique Vessels	DISTINCTCOUNT(Fact_VesselMovement[MMSI])	Counts the distinct vessels represented in the dataset.
Total Vessel Records	COUNTROWS(Fact_VesselMovement)	Counts the total number of vessel movement records.
Average Speed	AVERAGE(Fact_VesselMovement[SOG_kmh])	Calculates the average vessel speed in km/h.
Average ETA	AVERAGE(Fact_VesselMovement[ETA_hours])	Calculates the average estimated time of arrival in hours.
Average Distance	AVERAGE(Fact_VesselMovement[dist_km])	Calculates the average distance used in dashboard analysis.
DAX measures used to calculate the key dashboard indicators		

8. KPI DEVELOPMENT

After creating the required DAX calculations, the next step was to develop the Key Performance Indicators (KPIs) for the dashboard. KPIs were created to provide a quick and clear summary of important vessel movement information without requiring the user to examine individual records.

Four main KPI cards were included in the final dashboard:

8.1 TOTAL UNIQUE VESSELS

This KPI represents the number of distinct vessels available in the dataset. It provides a quick understanding of the overall number of individual vessels represented by the movement records.

8.2 TOTAL VESSEL RECORDS

This KPI represents the total number of vessel movement records available in the dataset. Since a single vessel can have multiple movement observations, this value provides an overview of the overall volume of recorded vessel activity.

8.3 AVERAGE SPEED

The Average Speed KPI provides the overall average movement speed of the vessels. It helps give a quick indication of the general movement condition represented in the dataset.

8.4 AVERAGE ETA

The Average ETA KPI represents the average estimated time of arrival based on the available vessel records. It provides a summarized view of the expected arrival information.

MARITIME KPI DASHBOARD			
KPI		What It Represents	Purpose
Total Unique Vessels		Number of distinct vessels	Provides an overview of vessel count
Total Vessel Records		Total movement observations	Shows the overall volume of vessel activity
Average Speed		Average vessel movement speed	Provides an overall view of vessel movement
Average ETA		Average estimated arrival time	Provides a summarized view of arrival information

9. DASHBOARD DEVELOPMENT

After completing data preparation, data modeling, DAX calculations, and KPI development, the final step of Milestone 1 was to develop an interactive Power BI dashboard. The dashboard was designed to present the important vessel movement information in a clear and organized manner so that users can understand the overall situation without going through the complete dataset manually.

The dashboard combines KPI cards, analytical charts, a detailed data table, and an interactive slicer. The layout was designed to keep the most important summary information at the top while providing more detailed analysis through the visualizations below.

9.1 DASHBOARD LAYOUT AND DESIGN

The dashboard was arranged into different sections according to the type of information being presented.

The top section contains the main KPI cards, which provide an immediate summary of the vessel movement data. The analytical charts are placed below the KPIs so that users can move from a high-level overview to more detailed analysis.

The dashboard was also given a consistent visual theme with a suitable background, colored KPI cards, and formatted visual elements. The objective was to make the dashboard both visually appealing and easy to read, while avoiding unnecessary visual clutter.

9.2 KPI CARDS

Four KPI cards were placed prominently on the dashboard:

- **Total Unique Vessels**
- **Total Vessel Records**
- **Average Speed**
- **Average ETA**

These cards provide the first level of information to the user. Instead of examining the complete dataset, users can immediately see the main numerical indicators.

The KPI values are generated using the DAX measures developed earlier, which allows them to respond to the data and applied filters.

9.3 ANALYTICAL VISUALIZATIONS

Several charts were included to analyze different aspects of vessel movement.

Speed Category Distribution

This visual shows the distribution of vessel movement records across different speed categories. It provides an easy way to understand how the recorded vessels are distributed according to their movement speed.

Average Distance by Status

This chart compares the average distance associated with different vessel status categories. It helps in understanding how distance varies according to the operational status of the vessels.

Average ETA by Hour

This visualization uses the time-related fields created during data preparation. It shows how the average estimated time of arrival changes across different hours.

This visual demonstrates the usefulness of the Hour and Hour_Label fields created during the data preparation stage.

Vessel Type Distribution

This chart provides an overview of the different vessel types represented in the dataset. It helps users understand the composition of the vessel records.

9.4 VESSEL DETAILS TABLE

A detailed table was included in the dashboard to complement the graphical visualizations.

While charts provide summarized information, the table allows users to inspect individual vessel records and their associated information. This provides a more detailed view when a user needs to examine specific records rather than only overall trends.

9.5 INTERACTIVE VESSEL STATUS SLICER

A Vessel Status slicer was included to make the dashboard interactive.

The slicer allows users to select a particular vessel status and analyze the dashboard according to that selection. When a status is selected, the relevant visualizations and KPI values can respond to the applied filter.

This makes the dashboard more useful than a static collection of charts because users can interact with the data and focus on the information that is relevant to them.

MILESTONE 2 - INVENTORY & DELIVERY ANALYTICS

10.1. INTRODUCTION

Following the completion of Milestone 1, the existing data model and KPI foundation were further extended in Milestone 2 to perform deeper operational analysis. This milestone focuses on analyzing vessel movement, destination, cargo, distance, ETA, and variance-related information using Power BI.

The dashboard was enhanced with additional KPI measures, analytical charts, drill-down analysis, trend and variance analysis, and interactive filtering. These enhancements provide deeper insights into operational patterns and support more detailed analysis of vessel movement and arrival performance.

10.2. MILESTONE 2 REQUIREMENTS & OBJECTIVES

10.2.1 REQUIREMENTS

The main requirements for Milestone 2 were to extend the Power BI dashboard with deeper inventory, delivery, drill-down, trend, and variance analysis. The suggested visualizations included:

- **KPI Cards** for important performance indicators.
- **Bar Chart** for category-wise analysis.
- **Column Chart** for warehouse or location-wise analysis.
- **Line Chart** for analyzing trends over time.
- **Table** for detailed record-level information.
- **Donut Chart** for comparing on-time and delayed deliveries.
- **Drill-down charts** for hierarchical analysis.
- **Trend and variance visuals** to identify changes and differences in performance.

10.2.2 OBJECTIVES

The objectives of Milestone 2 were:

1. To enhance the existing Power BI dashboard with deeper operational analysis.
2. To analyze vessel movement, destination, cargo, distance, and ETA-related performance.

3. To identify slow-moving and fast-moving vessel records using KPI measures.
4. To analyze movement and ETA trends across different time periods and destinations.
5. To identify variations in ETA performance using variance analysis.
6. To provide drill-down capability from Cargo to Vessel for detailed analysis.
7. To provide interactive filtering through the Destination Cluster slicer.
8. To present the analysis through clear and visually understandable Power BI charts.

10.3. DATA & EXISTING MODEL

Milestone 2 uses the existing data model developed during Milestone 1. The model is designed to organize vessel movement and date-related information for analysis in Power BI. No separate data model was created for Milestone 2; instead, the existing model was extended through additional DAX measures and analytical visuals.

10.3.1 DIMDATE TABLE

The **DimDate** table provides the date-related fields required for time-based analysis. It contains:

- Date
- Year
- Month
- Month Number
- Year-Month

The table is used to support date and time-based analysis and maintain consistency across the dashboard.

10.3.2 FACT_VESSELMOVEMENT TABLE

The **Fact_VesselMovement** table contains the main vessel movement and operational data used throughout the dashboard. Important fields used in Milestone 2 include:

Important Fields Used in Milestone 2

Field	Purpose
MMSI	Identifies vessel records
VesselName	Identifies individual vessels
Cargo	Supports cargo-wise analysis
dest_cluster	Supports destination-wise analysis
dist_km	Used for distance analysis
ETA_hours	Used for ETA and variance analysis
SOG_kmh	Used for vessel speed analysis
Speed_Category	Identifies movement speed categories

10.3.3 MODEL UTILIZATION IN MILESTONE 2

The existing model was used to create new analytical measures and visualizations for:

- Destination and distance analysis
- ETA analysis
- Vessel movement trends
- Cargo analysis
- Drill-down from **Cargo** → **Vessel**
- Variance analysis
- Distance and ETA relationship analysis

This allowed Milestone 2 to build upon the foundation created in Milestone 1 without duplicating the complete data preparation and modelling process.

10.4. DAX MEASURES DEVELOPMENT

To support the analytical requirements of Milestone 2, five new DAX measures were developed in Power BI using the existing Fact_VesselMovement table. These measures were used to create KPI cards and analytical visuals for distance, vessel speed, ETA variability, and destination-wise ETA analysis.

10.4.1 TOTAL DISTANCE TRAVELLED (KM)

Purpose:

This measure calculates the total recorded distance travelled by all vessel movement records.

DAX Formula:

Total Distance Travelled (km) =
SUM('Fact_VesselMovement'[dist_km])

Explanation:

The SUM function adds all the values from the dist_km column to calculate the overall distance travelled. The resulting value is used as an important KPI to understand the total movement distance represented in the dataset.

Used in:

- KPI Card – **Total Distance Travelled (km)**
- Destination-wise distance analysis

10.4.2 SLOW-MOVING VESSELS

Purpose:

This measure calculates the number of unique vessels classified as **Slow** according to the Speed_Category field.

DAX Formula:

Slow-Moving Vessels =
CALCULATE(
 DISTINCTCOUNT('Fact_VesselMovement'[MMSI]),
 'Fact_VesselMovement'[Speed_Category] = "Slow"
)

Explanation:

DISTINCTCOUNT counts each vessel only once using its MMSI. The CALCULATE function then applies a filter so that only records having the speed category "**Slow**" are considered.

This provides a KPI for identifying the number of unique slow-moving vessels in the dataset.

Used in:

- KPI Card – **Slow-Moving Vessels**

10.4.3 FAST-MOVING VESSELS

Purpose:

This measure calculates the number of unique vessels classified as **Fast**.

DAX Formula:

Fast-Moving Vessels =

```
CALCULATE(  
    DISTINCTCOUNT('Fact_VesselMovement'[MMSI]),  
    'Fact_VesselMovement'[Speed_Category] = "Fast"  
)
```

Explanation:

The measure uses DISTINCTCOUNT on the MMSI field to count unique vessels. CALCULATE filters the data so that only vessels with the **"Fast"** speed category are included.

This helps provide a quick view of the number of fast-moving vessels represented in the dataset.

Used in:

- KPI Card – **Fast-Moving Vessels**

10.4.4 ETA VARIABILITY (HRS)

Purpose:

This measure calculates the variability of the ETA values in hours.

DAX Formula:

```
ETA Variability (hrs) =  
STDEV.P('Fact_VesselMovement'[ETA_hours])
```

Explanation:

The STDEV.P function calculates the **population standard deviation** of the ETA_hours values. A higher value indicates greater variation in ETA values, while a lower value indicates that the ETA values are more consistent.

This measure was used to provide a high-level view of ETA variability in the dashboard.

Used in:

- KPI Card – **ETA Variability (hrs)**

10.4.5 ETA VARIANCE (HRS)

Purpose:

This measure calculates the difference between the current average ETA and the overall average ETA across all destination clusters.

DAX Formula:

ETA Variance (hrs) =

```
AVERAGE('Fact_VesselMovement'[ETA_hours])  
-  
CALCULATE(  
    AVERAGE('Fact_VesselMovement'[ETA_hours]),  
    ALL('Fact_VesselMovement'[dest_cluster])  
)
```

Explanation:

The first AVERAGE calculates the average ETA in the current filter context, such as a particular destination cluster.

The CALCULATE function calculates the overall average ETA while ALL('Fact_VesselMovement'[dest_cluster]) removes the destination-cluster filter.

The two values are then subtracted to determine how much the selected destination's average ETA differs from the overall average.

- **Positive value:** Average ETA is higher than the overall average.
- **Negative value:** Average ETA is lower than the overall average.
- **Value near zero:** Average ETA is close to the overall average.

Used in:

- **ETA Variance by Destination** chart

10.5. DASHBOARD DEVELOPMENT

For Milestone 2, an interactive Power BI dashboard was developed to provide deeper analysis of vessel movement and operational performance. The dashboard builds upon the data model and measures developed in the previous milestone and presents the results through KPI cards, charts, drill-down analysis, and interactive filtering.

The dashboard focuses on distance, vessel movement, destination, ETA, cargo, and variance analysis. The visuals were designed to provide both high-level performance indicators and detailed analytical insights while maintaining a clear and user-friendly layout.

Then we'll break the dashboard explanation into **four subsections**:

10.5.1 KPI ANALYSIS

Four KPI cards were developed to provide a quick overview of important operational indicators in the vessel movement dataset. These KPIs allow users to understand the overall movement distance, ETA variability, and vessel speed distribution at a glance.

1. ETA Variability (hrs)

The ETA Variability KPI represents the variation in estimated arrival times across the available vessel movement records. It is calculated using the population standard deviation of ETA_hours. This helps indicate how consistent or variable the ETA values are across the dataset.

2. Total Distance Travelled (km)

The Total Distance Travelled KPI shows the cumulative recorded distance covered by vessel movement records. It provides a high-level view of the overall movement distance represented in the dataset.

3. Slow-Moving Vessels

The Slow-Moving Vessels KPI represents the number of unique vessels classified under the Slow speed category. It helps identify the portion of vessel activity associated with slower movement.

4. Fast-Moving Vessels

The Fast-Moving Vessels KPI represents the number of unique vessels classified under the Fast speed category. It provides a quick indication of vessels operating at higher movement speeds.

10.5.2 DESTINATION & MOVEMENT ANALYSIS

This section explains the visuals used to analyze destination-wise vessel activity, movement patterns over time, and ETA performance.

1. Total Distance by Destination

The Total Distance by Destination chart compares the total recorded vessel movement distance across different destination clusters. It helps identify destinations associated with higher or lower overall vessel movement distance and provides a clearer view of destination-wise operational activity.

2. Vessel Movement Trend by Hour

The Vessel Movement Trend by Hour line chart shows how vessel movement activity changes throughout the day. The hourly analysis helps identify periods of relatively higher and lower vessel activity, making it useful for understanding the time-based pattern of vessel movements.

3. Average ETA by Destination

The Average ETA by Destination chart compares the average estimated arrival time across destination clusters. It helps identify destinations with relatively higher or lower average ETA values and provides a basis for comparing arrival-time performance between different destinations.

10.5.3 TREND, VARIANCE & RELATIONSHIP ANALYSIS

This part of the dashboard focuses on identifying differences in ETA performance and understanding the relationship between distance, ETA, and vessel speed.

1. ETA Variance by Destination

The ETA Variance by Destination chart compares the average ETA of each destination cluster with the overall average ETA. The variance measure helps identify destinations where the average ETA differs from the overall benchmark.

- A **positive variance** indicates that the destination's average ETA is higher than the overall average.
- A **negative variance** indicates that the destination's average ETA is lower than the overall average.
- A value **close to zero** indicates that the destination's average ETA is close to the overall average.

This visualization helps users identify destinations with noticeable deviations in ETA performance.

2. Distance vs ETA Analysis

The Distance vs ETA Analysis scatter chart is used to examine the relationship between vessel movement distance and estimated arrival time across different speed categories.

The visual helps in understanding whether vessels associated with greater distances also tend to have higher ETA values, while the speed-category breakdown provides additional context about vessel movement behavior.

10.5.4 CARGO & DRILL-DOWN ANALYSIS

This section focuses on cargo-wise movement analysis and detailed vessel-level exploration. It also includes the interactive destination filter used to make the dashboard more dynamic.

1. Top 10 Cargo Categories by Average Distance

The Top 10 Cargo Categories by Average Distance bar chart compares cargo categories based on their average recorded movement distance. The visual is limited to the top 10 categories to keep the analysis clear and readable.

This helps identify cargo categories associated with relatively higher average movement distances and provides an additional perspective for understanding cargo-related vessel activity.

2. Cargo → Vessel Drill-Down

The Cargo → Vessel Drill-Down chart provides hierarchical analysis by allowing users to move from a high-level Cargo category to individual VesselName records.

This enables users to:

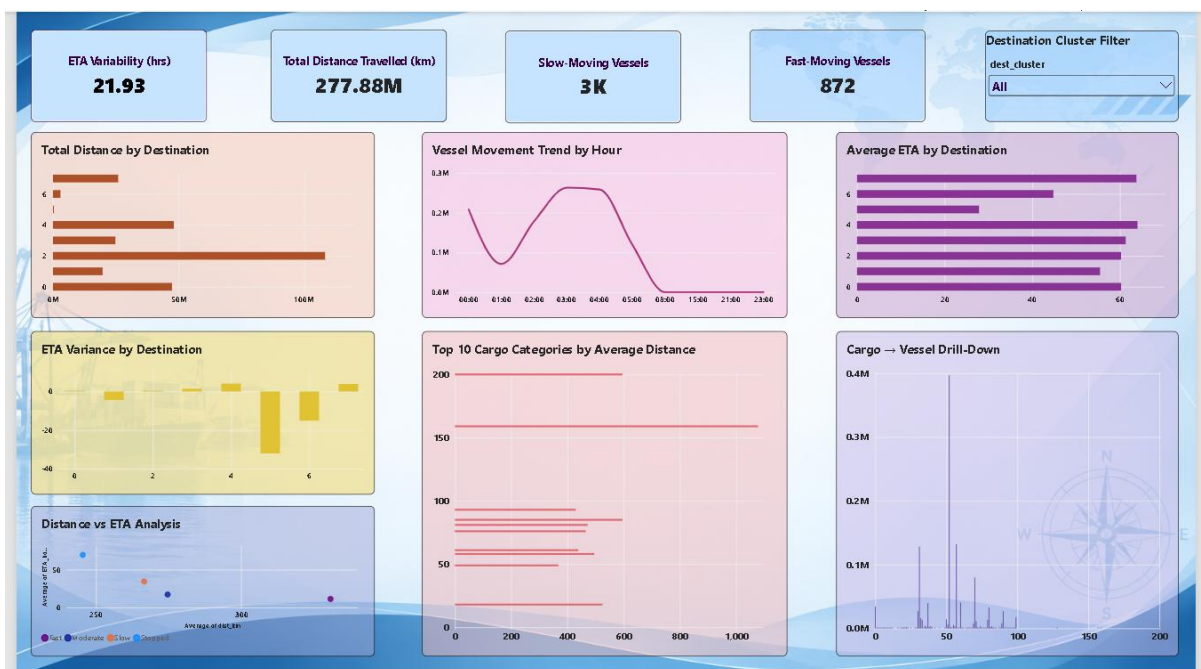
- Analyze cargo categories at a summarized level.
- Select a cargo category for further investigation.
- Drill down to the vessels associated with the selected cargo.
- Examine vessel-level details within a particular cargo category.

This fulfills the drill-down analysis requirement of Milestone 2 while providing more detailed exploration of the available data.

3. Destination Cluster Filter

A Destination Cluster dropdown slicer was added to the dashboard to provide interactive filtering. Users can select a particular destination cluster and dynamically update the dashboard visuals according to the selected destination.

This allows the analysis to be performed both at an overall level and for a specific destination cluster.



MILESTONE 3 - SUPPLIER & TRANSPORTATION ANALYTICS

3.1 DATA AND MODEL ADAPTATION

For Milestone 3, the existing vessel movement dataset and Power BI data model were extended to support transportation and route performance analysis. The source AIS dataset contains fields such as vessel type, cargo, destination cluster, movement distance, speed, status, and MMSI. However, it does not contain direct business fields such as supplier, carrier, actual transportation cost, shipment cost, supplier quality score, fulfillment status, or actual delivery time.

To address this limitation, the available fields were used to develop relevant transportation and route performance metrics. A configurable Cost per KM What-if parameter was also created to estimate transportation cost from the available dist_km field. The parameter allows the estimated cost to change dynamically based on the selected cost-per-kilometer value.

The existing DimDate and Fact_VesselMovement relationship was retained, while the new Cost per KM parameter table was kept disconnected from the main fact table because it is used as a user-controlled parameter rather than as a data dimension.

3.2 COST PER KM PARAMETER DEVELOPMENT

Since the source dataset does not contain actual transportation cost or shipment cost information, a What-if parameter named Cost per KM was created in Power BI. This parameter provides a configurable cost assumption that can be used with the available vessel movement distance to calculate estimated transportation cost.

The parameter was configured with the following values:

- **Parameter Name:** Cost per KM
- **Data Type:** Decimal Number
- **Minimum Value:** 10
- **Maximum Value:** 100
- **Increment:** 5
- **Default Value:** 50

The parameter generates a corresponding Cost per KM Value measure. This measure is used in the transportation-cost calculation. When the user changes the Cost per KM value through the dashboard slider, the estimated transportation cost and related visuals update dynamically.

The parameter table was intentionally kept disconnected from the Fact_VesselMovement table because it is a user-controlled calculation parameter and does not represent a separate business entity in the source dataset.

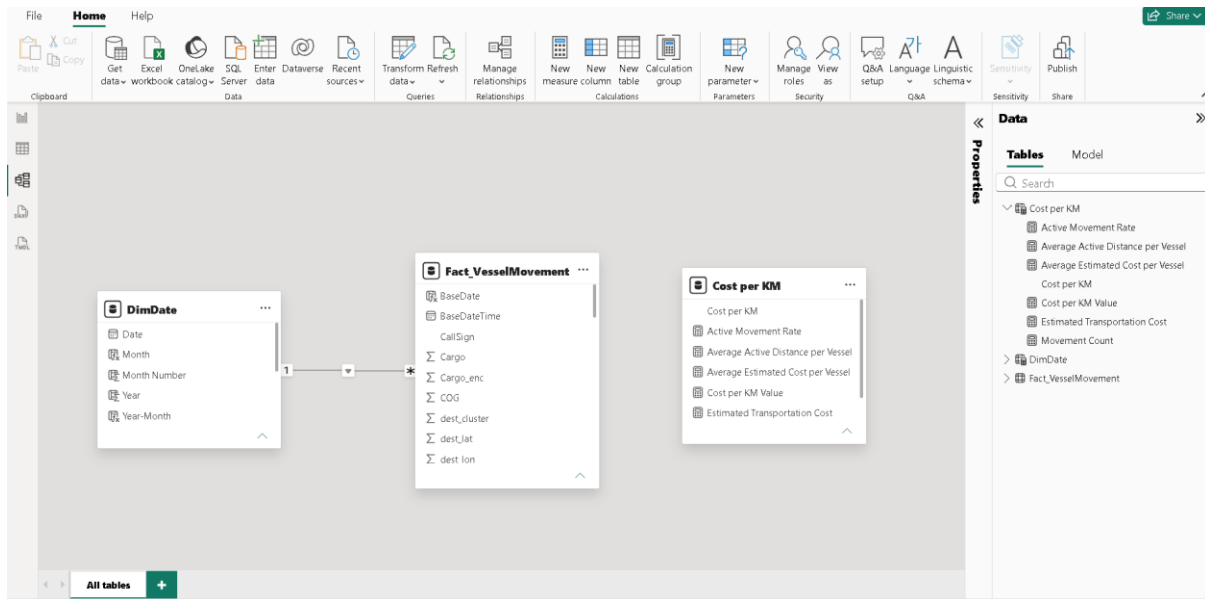


Figure 3.1: M3 Power BI Data Model and Cost per KM Parameter

3.3 DAX MEASURE DEVELOPMENT

To support the transportation and route performance analysis, several DAX measures were created in Power BI. These measures use the existing vessel movement data along with the newly created Cost per KM parameter.

3.3.1 ESTIMATED TRANSPORTATION COST

The **dist_km** field was used with the selected Cost per KM value to calculate the estimated transportation cost.

Estimated Transportation Cost =

`SUM('Fact_VesselMovement'[dist_km]) * [Cost per KM Value]`

This measure dynamically changes when the Cost per KM assumption is changed using the dashboard control.

3.3.2 AVERAGE ESTIMATED COST PER VESSEL

This measure calculates the average estimated transportation cost per unique vessel using MMSI.

Average Estimated Cost per Vessel =

`DIVIDE(`

`[Estimated Transportation Cost],`

`DISTINCTCOUNT('Fact_VesselMovement'[MMSI])`

)

3.3.3 ACTIVE MOVEMENT RATE

Since the dataset does not contain an actual fulfillment or delivery-completion field, the movement status was used to create an operational activity metric. Records where the vessel status is not Stopped are treated as active movement records.

Active Movement Rate =

```
DIVIDE(  
  CALCULATE(  
    COUNTROWS('Fact_VesselMovement'),  
    'Fact_VesselMovement'[Speed_Category] <> "Stopped"  
  ),  
  COUNTROWS('Fact_VesselMovement')  
)
```

The resulting percentage represents the proportion of movement records classified as active.

3.3.4 AVERAGE ACTIVE DISTANCE PER VESSEL

This measure calculates the average distance associated with vessels that were actively moving.

Average Active Distance per Vessel =

```
DIVIDE(  
  CALCULATE(  
    SUM('Fact_VesselMovement'[dist_km]),  
    'Fact_VesselMovement'[Speed_Category] <> "Stopped"  
  ),  
  CALCULATE(  
    DISTINCTCOUNT('Fact_VesselMovement'[MMSI]),  
    'Fact_VesselMovement'[Speed_Category] <> "Stopped"  
  )  
)
```

This metric helps analyse the average distance associated with active vessel movement.

3.3.5 MOVEMENT COUNT

A movement-count measure was created to support the scatter chart and represent the volume of movement records.

Movement Count =

COUNTROWS('Fact_VesselMovement')

This measure was used as the **Size** field in the Transportation Cost vs Average Speed scatter chart.

3.3.6 MEASURE VALIDATION

After creating the measures, they were added to the M3 dashboard and validated through KPI cards and analytical visuals. The Cost per KM parameter was also tested by changing its value and observing the corresponding changes in the estimated transportation-cost measures.

3.4 DASHBOARD DEVELOPMENT

The M3 dashboard was developed as a new report page in the existing Power BI file. The dashboard focuses on estimated transportation cost, vessel movement performance, destination-level cost analysis, and interactive parameter-based analysis.

The dashboard was designed using KPI cards, analytical charts, a performance scorecard, interactive slicers, and a configurable Cost per KM control.

3.4.1 KPI CARDS

Four KPI cards were created to provide a quick overview of transportation cost and vessel movement performance. These cards help users understand the main values without checking every individual chart.

1. Estimated Transportation Cost

The Estimated Transportation Cost card displays the total estimated cost of vessel movement records. It is calculated by multiplying the total recorded movement distance by the selected Cost per KM value.

At the default Cost per KM assumption of 50, the dashboard displays approximately **\$13.89 billion**.

This value is an estimated analytical value and does not represent actual transportation invoices or financial expenditure.

2. Average Estimated Cost per Vessel

The Average Estimated Cost per Vessel card shows the average estimated transportation cost associated with each unique vessel. The calculation uses the total estimated transportation cost divided by the distinct number of vessels identified through the MMSI field.

The displayed value is approximately **\$1.09 million per vessel** at the default Cost per KM assumption.

This KPI helps provide a vessel-level view of the estimated transportation cost.

3. Active Movement Rate

The Active Movement Rate card shows the percentage of movement records in which the vessel was not classified as stopped.

Because the dataset does not contain actual order fulfillment or delivery-completion information, this metric was created as an operational activity indicator using the Speed_Category field.

The dashboard displays an active movement rate of approximately **20.36%**.

This value represents movement-record activity, not an actual order fulfillment rate or delivery success rate.

4. Average Active Distance per Vessel

The Average Active Distance per Vessel card shows the average recorded distance associated with vessels that were actively moving. Records classified as stopped are excluded from this calculation.

The displayed value is approximately **18.61K km**.

This KPI helps analyse the average movement distance of active vessels and provides additional operational context for the transportation-cost analysis.

3.4.2 ESTIMATED TRANSPORTATION COST BY VESSEL TYPE

A horizontal bar chart was created to compare the estimated transportation cost associated with different vessel types.

The chart uses the VesselType field as the category and the Estimated Transportation Cost measure as the value. The vessel types are sorted in descending order so that the categories with higher estimated transportation costs appear first.

The horizontal bar format was selected because the dataset contains multiple vessel types. Compared with a vertical column chart, horizontal bars provide more space for displaying vessel-type names and make category comparisons easier to read.

This visual helps identify how estimated transportation cost is distributed across different vessel categories. It also allows users to compare vessel types based on the selected Cost per KM assumption.

The chart is interactive and responds to the Vessel Type, Cargo, Destination Cluster, and Cost per KM controls available on the M3 dashboard.

The visual does not represent actual carrier-wise transportation expenditure because the source dataset does not contain carrier or financial cost fields. Instead, it provides a vessel-type-based estimate using the available movement-distance data.

3.4.3 ESTIMATED TRANSPORTATION COST BY DESTINATION CLUSTER

A donut chart was created to analyse the distribution of estimated transportation cost across the different destination clusters in the dataset.

The chart uses `dest_cluster` as the category and Estimated Transportation Cost as the value. Since the dataset contains **8 destination clusters**, the donut chart provides a compact view of how the estimated transportation cost is distributed among these destinations.

The visual changes dynamically according to the selected Cost per KM Assumption. It also responds to the other dashboard filters, allowing the cost distribution to be analysed for specific vessel types, cargo categories, or destination clusters.

This analysis provides a destination-level perspective of transportation cost and complements the vessel-type cost analysis presented in the previous visual.

The chart represents estimated transportation cost, calculated using the available movement-distance data and the configurable Cost per KM assumption. It does not represent actual financial transportation expenditure because the source dataset does not contain actual cost records.

3.4.4 VESSEL TYPE PERFORMANCE SCORECARD

A matrix-based performance scorecard was created to provide a detailed comparison of different vessel types using multiple transportation and movement metrics.

The VesselType field is used as the row category, while the following measures are displayed:

- **Estimated Transportation Cost**
- **Active Movement Rate**
- **Average Active Distance per Vessel**

The scorecard allows multiple performance indicators to be viewed together rather than analysing each metric separately. It is sorted according to estimated transportation cost to make comparison between vessel types easier.

This visual was developed as an adaptation of the supplier-performance concept suggested for the milestone. Since the source dataset does not contain supplier identifiers or supplier quality

scores, actual supplier-wise quality analysis could not be performed. Instead, the available vessel-level information was used to create a performance-oriented scorecard.

3.4.5 TRANSPORTATION COST VS AVERAGE SPEED

A scatter chart was created to analyse the relationship between vessel movement speed and estimated transportation cost. This visual helps compare vessel types based on their average speed and the estimated transportation cost associated with their movement records.

The visual was configured using the following fields:

- **X-Axis:** Average Speed
- **Y-Axis:** Estimated Transportation Cost
- **Legend:** VesselType
- **Size:** Movement Count
- **Filter:** Top 10 Vessel Types by Estimated Transportation Cost

The Top 10 filter was applied to reduce visual clutter because the dataset contains a large number of vessel types. The bubble size represents the number of movement records, allowing the user to see both cost and movement volume in the same visual.

The chart responds dynamically to the dashboard slicers and the Cost per KM assumption. Since transportation cost is estimated from movement distance and the selected cost-per-kilometer value, changing the parameter also changes the position of the bubbles on the cost axis.

This analysis provides an operational comparison between vessel speed, movement volume, and estimated transportation cost. It should be interpreted as an analytical relationship within the AIS movement dataset rather than as an actual financial cost-versus-speed analysis.

3.4.6 INTERACTIVE SLICERS AND COST ASSUMPTION CONTROL

To make the M3 dashboard interactive, four slicers were added to allow users to filter and analyse the transportation data based on different operational factors.

The slicers used in the dashboard are:

1. **Vessel Type**
Allows users to focus the analysis on a particular type of vessel.
2. **Cargo**
Allows the dashboard to be filtered according to the available cargo categories.
3. **Destination Cluster**
Allows users to analyse transportation cost and movement performance for a selected destination cluster.

4. Cost per KM Assumption (\$)

This is a What-if parameter control that allows users to change the assumed transportation cost per kilometre. The available range is **\$10 to \$100**, with increments of **\$5**.

The categorical slicers help users explore specific portions of the dataset, while the Cost per KM control changes the estimated transportation cost dynamically. When the assumption is changed, the related KPI cards and cost-based visuals are recalculated automatically.

The Cost per KM control is an analytical assumption rather than an actual business field from the source dataset. Therefore, the resulting transportation cost should be interpreted as an estimated value based on the selected assumption.

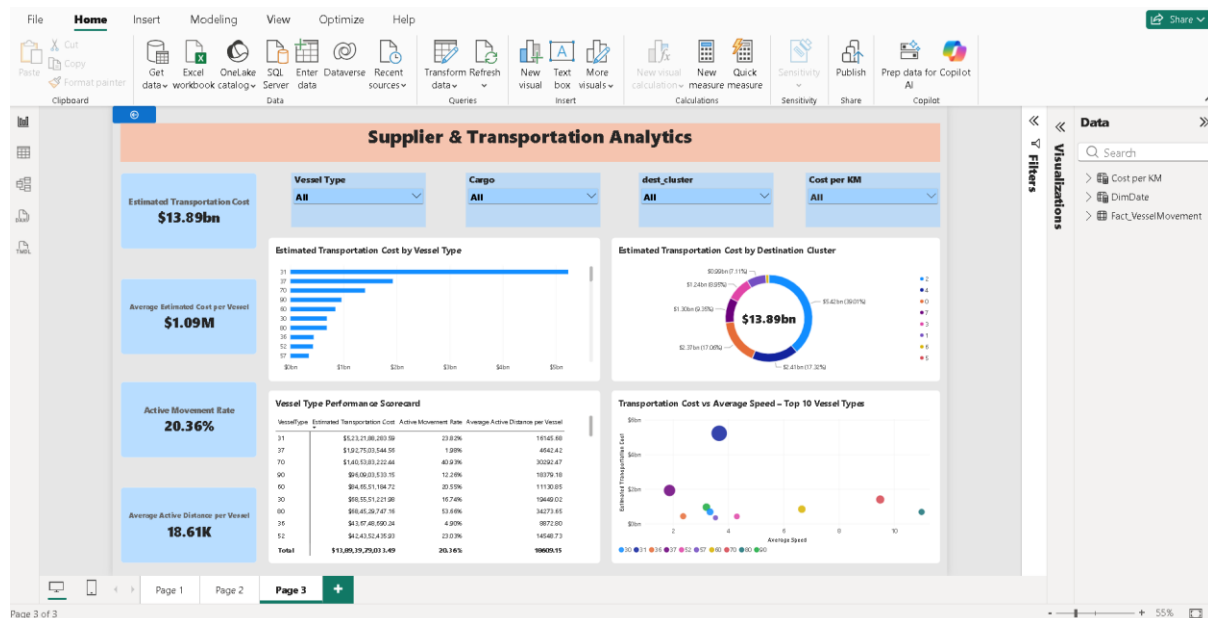


Figure 3.2: Final Milestone 3 Dashboard Output

MILESTONE 4 – WAREHOUSE ANALYTICS & FINAL DASHBOARD

1. INTRODUCTION AND IMPLEMENTATION OVERVIEW

The fourth milestone focuses on developing the final Power BI dashboard, including warehouse efficiency analysis, an executive overview, dashboard optimization and final validation.

The dashboard consists of three pages: Warehouse Efficiency & Zone Analytics, Supply Chain Executive Overview, and Dashboard Performance & Validation. These pages present key performance indicators, vessel movement patterns, destination-zone comparisons and estimated transportation costs.

Since the AIS dataset does not contain actual warehouse data, destination clusters are used as proxy operational zones. Transportation costs are estimated using an assumed cost per kilometre.

The existing data model and DAX measures were reused and extended to complete the final dashboard and improve its readability and analytical capabilities.

2. WAREHOUSE EFFICIENCY & ZONE ANALYTICS

The Warehouse Efficiency & Zone Analytics dashboard provides an overview of vessel movement activity across different destination zones. Since the source dataset contains AIS vessel movement information rather than actual warehouse records, destination clusters are used as proxy operational zones for analysis. The dashboard brings together key performance indicators (KPIs) and visualizations to examine movement activity, vessel behavior, estimated arrival time (ETA) variability, and differences between destination zones.

The page is designed to help users identify zones with higher movement activity, compare active movement rates, and observe variations in vessel speed and estimated arrival times. These insights provide an analytical view of vessel movement patterns, while avoiding the assumption that the destination clusters represent physical warehouses.

2.1 KEY PERFORMANCE INDICATORS

1. Total Vessel Records

This KPI displays the total number of vessel movement records available in the dataset. Each record represents a vessel movement observation captured in the AIS data. It provides an overall measure of the volume of movement information being analyzed and serves as a reference point for interpreting the other KPIs and visualizations. When dashboard filters are applied, the value changes according to the selected date, vessel speed, status, or other relevant filtering conditions.

2. Active Movement Rate

The Active Movement Rate represents the percentage of vessel movement records classified as active, excluding records categorized as stopped. It is calculated by dividing the number of active movement records by the total number of movement records. This KPI helps assess the proportion of observed vessel activity that involves movement rather than stopped status. Comparing this rate across destination zones can help identify differences in vessel activity patterns. It should be interpreted as an AIS movement indicator, not as a measure of warehouse utilization or operational efficiency.

3. Total Destination Zones

This KPI displays the number of distinct destination clusters present in the dataset. These clusters are treated as proxy operational zones to support comparative analysis of vessel movements. The measure helps users understand how many destination areas are represented in the available data and provides context for the zone-based charts on this page. The clusters are dataset-derived categories and should not be interpreted as verified warehouse locations.

4. ETA Variability

ETA Variability measures the statistical variation in estimated arrival times, using the standard deviation of the ETA values. It provides an indication of how widely vessel arrival estimates differ within the selected data. A higher value indicates greater dispersion in estimated arrival times, while a lower value indicates that the ETA values are more closely grouped. This KPI supports the analysis of arrival-time consistency across the dataset. It reflects variation in the available ETA estimates and does not independently establish whether vessels arrived early, late, or on schedule.

5. Average Active Distance per Vessel

This KPI calculates the average recorded travel distance per distinct vessel among movement records classified as active. It is obtained by dividing the total distance associated with active movement records by the number of distinct vessels with active records. The measure provides an overview of the distance covered by active vessels in the dataset and can be used alongside the active movement rate to understand movement patterns. Since it is based on the available AIS observations, it should be interpreted as a dataset-level movement measure rather than a direct measure of warehouse or transportation productivity.

2.2 DASHBOARD VISUALIZATIONS

1. Movement Records by Destination Zone

This horizontal bar chart compares the number of vessel movement records associated with each destination zone. Each bar represents a destination cluster, and its length indicates the number of records recorded for that zone. The visualization makes it easier to identify zones with relatively high or low movement activity and compare their contribution to the overall dataset. It also helps users recognize how movement records are distributed across the proxy operational zones. A higher record count indicates more observations in the dataset, but does not necessarily mean that a physical warehouse is busier or has greater capacity.

2. Active Movement Rate by Destination Zone

This column chart displays the percentage of active vessel movement records for each destination zone. By comparing the columns, users can identify differences in the proportion of active movements across zones. The visualization complements the overall Active Movement Rate KPI by providing a zone-level breakdown rather than a single aggregated value. It can help highlight zones where a larger or smaller share of recorded vessel movements is classified as active. The result is dependent on the AIS movement classifications and should not be treated as a direct warehouse activity or utilization metric.

3. ETA Variability by Destination Zone

This horizontal bar chart compares ETA variability across destination zones using the standard deviation of estimated arrival times. It helps identify zones where the available ETA values show greater variation and zones where they are more closely grouped. The chart provides additional detail beyond the overall ETA Variability KPI by showing how the measure differs between clusters. Users can compare the relative spread of ETA estimates and investigate zones that may warrant further examination. The chart describes variability in estimated arrival times; it does not by itself measure actual delivery delays or on-time performance.

4. Top 5 Vessel Types by Movement Count

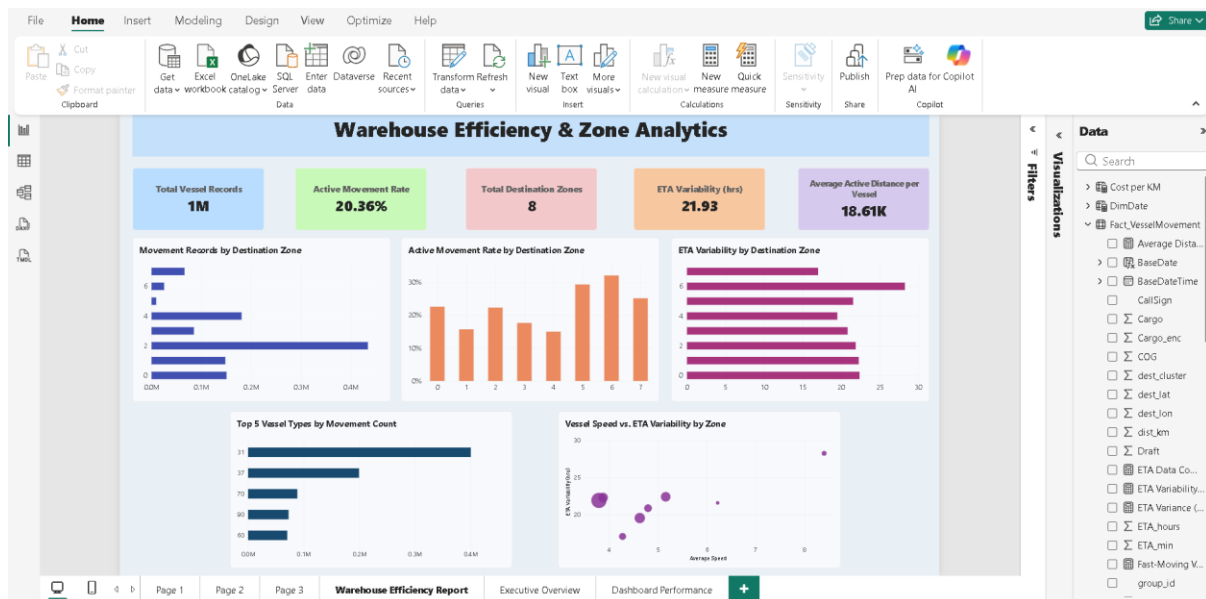
This bar chart displays the five vessel types with the highest number of movement records in the dataset. Limiting the visual to the top five categories keeps it readable and allows users to focus on the most frequently observed vessel types. The chart helps show which vessel categories contribute the largest share of recorded movement activity. It can also be used with dashboard filters to examine how the distribution changes for selected movement statuses, speeds, or destination zones. The ranking is based on record count, so it represents the frequency of observations rather than the number of physical vessels or their carrying capacity.

5. Vessel Speed vs. ETA Variability by Zone

This scatter chart compares vessel speed with ETA variability across destination zones. The horizontal axis represents the selected speed measure, while the vertical axis represents ETA variability. Each plotted point represents a zone-level comparison, allowing users to observe whether zones with different average vessel speeds also exhibit different levels of variation in estimated arrival times. The chart provides a compact way to explore possible relationships between movement speed and ETA patterns. However, any visible association should be treated as descriptive rather than proof that speed causes changes in ETA variability. The analysis is limited to the observations and fields available in the AIS dataset.

The completed Warehouse Efficiency & Zone Analytics dashboard combines the five KPI cards and five visualizations into a single analytical page. The KPI cards provide a high-level summary of movement records, active movement, destination coverage, ETA variability, and active-vessel distance. The charts then provide more detailed comparisons across destination zones and vessel types.

2.3 DASHBOARD SCREENSHOT



3. SUPPLY CHAIN EXECUTIVE OVERVIEW

The Supply Chain Executive Overview dashboard provides a consolidated view of vessel movement activity, estimated transportation costs, cargo distribution, and arrival-time patterns. It is designed to help users understand the overall movement status and review important operational indicators through a single interactive dashboard.

The page combines five KPI cards with four visualizations and interactive slicers. The KPIs summarize the key measures, while the charts provide further breakdowns by movement status, cargo category, vessel type, and destination zone. The dashboard uses AIS vessel movement data. Therefore, the estimated transportation cost is calculated using an assumed cost-per-kilometre value, and the destination clusters represent proxy operational zones rather than verified warehouses.

3.1 KEY PERFORMANCE INDICATORS

1. Total Unique Vessels

This KPI displays the number of distinct vessels identified in the dataset, based on the vessel identification field (MMSI). Unlike the total movement records KPI, which counts individual observations, this measure counts each vessel only once within the current filter context. It provides an overview of the number of unique vessels represented in the selected data. Users can apply dashboard filters to examine the distinct vessel count for particular dates, movement statuses, speeds, or other available categories.

2. Total Distance Travelled

This KPI represents the sum of the recorded vessel movement distances in kilometres. It provides an overall measure of the movement distance captured in the dataset and helps users understand the scale of vessel activity within the selected period or filter context. The value is calculated by aggregating the available distance field across the filtered movement records. Since the measure is based on the source AIS observations, it represents recorded movement distance rather than a verified end-to-end shipping route distance.

3. Average ETA

The Average ETA KPI displays the mean estimated arrival time, expressed in hours, for the vessel movement records included in the current selection. It provides a summary of the expected arrival duration represented in the dataset. This value can be examined alongside destination-zone and movement-status visualizations to explore differences in estimated arrival times across categories. It is an average of the available ETA estimates and should not be interpreted as a measure of actual arrival punctuality, since actual arrival timestamps and verified delivery commitments are not established by this dataset.

4. Estimated Transportation Cost

This KPI displays the estimated transportation cost calculated from recorded movement distance and an assumed cost-per-kilometre value. The calculation multiplies the total recorded distance by the selected cost-per-kilometre parameter. It provides a consistent illustrative cost estimate that can be explored through the dashboard's filters and zone-level breakdowns. Since the original AIS dataset does not include actual transportation expenses, invoices, or shipment-cost records, this KPI is an analytical estimate and must not be presented as actual business expenditure.

5. ETA Data Completeness

This KPI measures the proportion of movement records that contain a non-blank ETA value. It is calculated by dividing the number of records with an available ETA by the total number of movement records in the current selection. The measure helps users assess whether the ETA field is populated across the dataset and provides a simple data-quality indicator for the arrival-time analysis. A value of 100% indicates that all records in the current filter context contain an ETA value; it does not indicate that the ETA estimates are accurate or that vessels arrived on time.

3.2 DASHBOARD VISUALIZATIONS

1. Vessel Movement Status Distribution

This pie chart presents the distribution of vessel movement records across the available movement-status categories. Each slice represents a status category, and its size reflects the number or proportion of records belonging to that category. The visualization provides a quick overview of how the recorded movements are distributed among the different statuses. It complements the Total Unique Vessels and movement-related KPIs by showing the composition of the selected movement records. The chart describes the status values available

in the AIS dataset and does not independently establish operational performance or delivery success.

2. Top Cargo Categories by Movement Count

This treemap visualizes the leading cargo categories based on their vessel movement record counts. Each rectangle represents a cargo category, and its relative size reflects the number of movement records associated with that category. Applying a Top N filter helps focus the chart on the most frequently represented categories and improves readability when the dataset contains many cargo values. This visualization helps users identify the cargo categories that account for a larger share of recorded vessel activity. The measure represents movement-record frequency, not cargo weight, volume, monetary value, or actual inventory levels.

3. Movement Status by Top 5 Vessel Types

This stacked column chart compares movement-status records across the five vessel types with the highest movement counts. Each column represents a vessel type, while the stacked segments show the distribution of movement records across the available status categories. The chart enables users to compare both the total number of records and the status composition of each selected vessel type. Restricting the display to the top five vessel types reduces visual clutter and makes comparisons easier. The analysis is based on the vessel-type and status classifications in the AIS dataset; it should not be interpreted as a measure of vessel capacity, productivity, or service quality.

4. Average ETA by Destination Zone

This horizontal bar chart compares the average estimated arrival time across the destination clusters. Each bar represents a destination zone, and its length corresponds to the average ETA in hours for the records associated with that zone. Sorting the chart by average ETA makes it easier to identify zones with relatively higher or lower estimated arrival durations. The visualization provides a destination-level breakdown of the overall Average ETA KPI and supports further exploration using the dashboard filters. Since destination clusters are proxy operational zones and the dataset does not establish verified delivery schedules or actual arrival outcomes, the chart should be interpreted as a comparison of ETA estimates rather than a ranking of real-world delivery performance.

3.3 INTERACTIVE FILTERS

The dashboard includes slicers to allow users to explore the data from different perspectives.

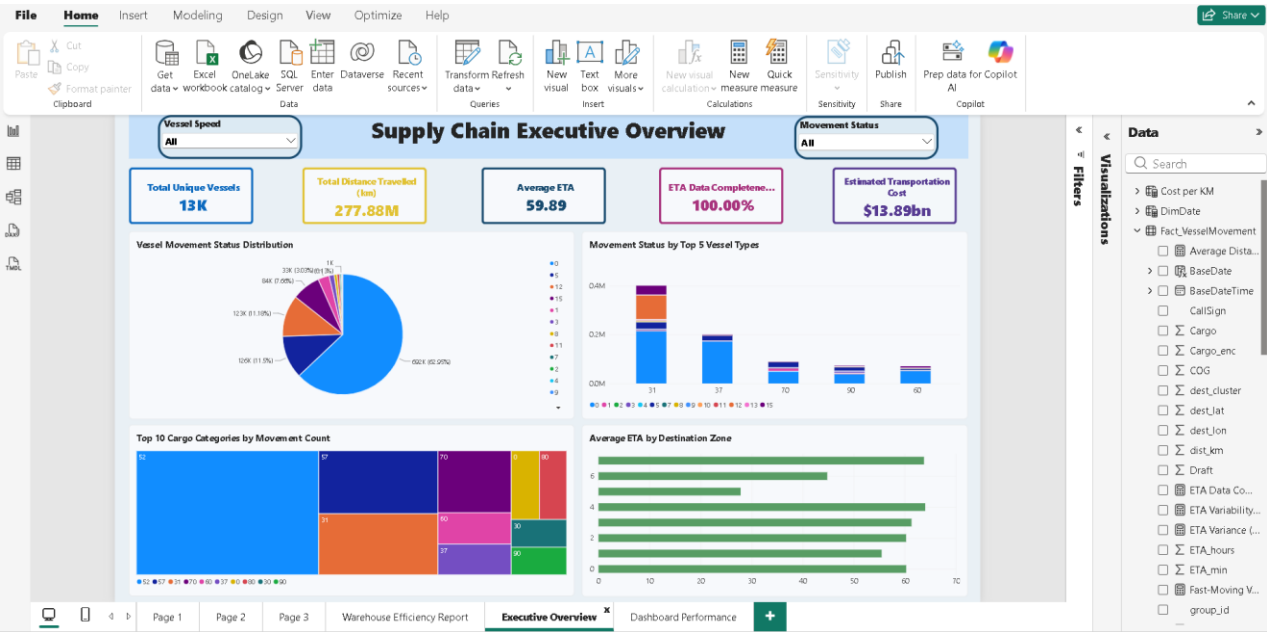
Date: Allows users to filter the dashboard by the available analysis date field. This helps narrow the displayed metrics and visualizations to the selected date or date range. The date slicer should use the related date dimension where applicable, and its behavior should be verified against the model's date relationship.

Vessel Speed: Allows users to filter the analysis by vessel speed category. This supports comparisons between different movement-speed classifications and helps examine how the selected category affects the KPIs and visualizations.

Movement Status: Allows users to select specific movement-status categories. This makes it possible to focus on particular types of recorded vessel activity and observe how the selection changes the dashboard results.

These slicers make the executive overview interactive and allow users to investigate the available AIS data without creating separate reports for every category or selection.

3.4 DASHBOARD SCREENSHOT



4. DASHBOARD PERFORMANCE & VALIDATION

4.1 DASHBOARD PERFORMANCE & VALIDATION TABLE

This page presents a single table summarizing vessel movement performance across destination zones. It allows users to compare key measures and review overall results in one place.

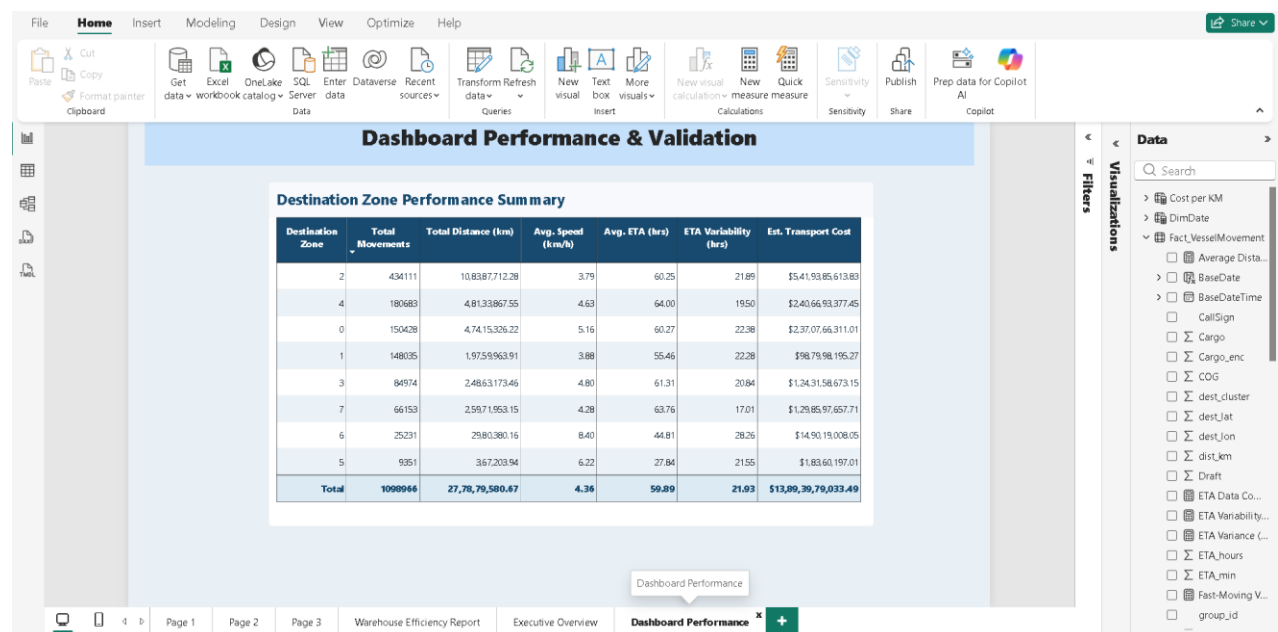
The table includes the following fields:

- 1. **Destination Zone:** Identifies each destination cluster, used as a proxy operational zone for comparing vessel movements.
- 2. **Movement Count:** Shows the number of vessel movement records associated with each zone.
- 3. **Total Distance:** Displays the total recorded vessel movement distance in kilometres for each zone.

4. **Average Speed:** Shows the average recorded vessel speed, helping compare movement patterns between zones.
5. **Average ETA:** Displays the average estimated arrival time in hours for each zone.
6. **ETA Variability:** Shows the variation in ETA values, helping identify differences in estimated arrival patterns.
7. **Estimated Cost:** Presents the calculated transportation cost based on recorded distance and an assumed cost-per-kilometre value. It is an estimate, not actual expenditure.

The table uses a dark blue header, alternating row colors, adjusted column widths, and a highlighted total row to improve readability.

4.2 DASHBOARD SCREENSHOT



4.3 DASHBOARD TESTING AND OPTIMIZATION

The completed dashboard pages were reviewed for measure accuracy, visual clarity, filter behavior, and consistent formatting. Existing DAX measures were reused where appropriate, and Top N filters were applied to selected charts to reduce clutter.

The report layout was organized to support easy navigation and comparison across pages. Power BI Performance Analyzer can be used to identify slow visuals and review loading times. Any performance improvement should be reported only if it has been measured.

The dashboard also clearly identifies destination clusters as proxy zones and transportation costs as estimates, since the source AIS dataset does not contain actual warehouse records or transportation expenses.

5. CONCLUSION

The Supply Chain Visibility System with Optimization Analytics project was successfully developed using Power BI and AIS vessel movement data. The project was completed across four milestones, covering data modelling, KPI development, inventory and delivery analytics, supplier and transportation analytics, and warehouse-zone analysis with an executive overview.

In Milestone 1, the foundation was established through data preparation, data modelling, and the development of essential vessel movement KPIs and visualizations. Milestone 2 expanded the analysis with inventory and delivery-related insights adapted to the available AIS dataset. Milestone 3 introduced supplier and transportation analytics concepts, including estimated transportation costs and movement-related measures. Milestone 4 completed the project with destination-zone analysis, an executive overview, and dashboard validation and performance considerations.

Throughout the project, Power BI, Power Query, and DAX were used to transform data, create relationships, develop measures, and build interactive dashboards. The project provided practical experience in data analysis, visualization, KPI development, and report design.

Although the source dataset does not contain actual warehouse, supplier, carrier, or transportation-cost records, the project demonstrates how available AIS data can be adapted for supply chain visibility analysis. Destination clusters were used as proxy operational zones, and transportation costs were estimated using an assumed cost-per-kilometre value.

Overall, the project strengthened practical skills in business intelligence and data analytics while demonstrating the development of a multi-page Power BI reporting solution through a structured, milestone-based approach.

6. REFERENCES

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